Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Fall, Year:2026), B.Sc. in CSE (Day)

Lab Report NO : 01
Course Title: Data Mining Lab
Course Code: CSE 436 Section: Old_Batch

Lab Experiment Name: Implementation of Data Preprocessing, Visualization, and Dimensionality Reduction Techniques in Python.

Student Details

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Lab Date : 26-10-2026
Submission Date : 02-11-2026
Course Teacher's Name : Md. Shoab Alam

Lab Report Status

Marks: ……. **Signature:.....**
Comments:..... **Date:.....**

1. TITLE OF THE LAB REPORT EXPERIMENT

Applying Python to Execute Data Preprocessing, Graphical Visualization, and Dimensionality Reduction Methods.

2. OBJECTIVES

- To understand data preprocessing techniques including binning (discretization), standardization, and categorical encoding.
- To handle missing data using imputation (mean/mode) and dropping techniques (dropna).
- To visualize data distributions and relationships using Seaborn (Bar plot, Histplot, Pairplot).
- To apply Principal Component Analysis (PCA) for linear dimensionality reduction and analyze explained variance.
- To apply t-SNE for non-linear dimensionality reduction and compare its clustering behavior with PCA.

3. PROCEDURE

❖ Step of The Problem:

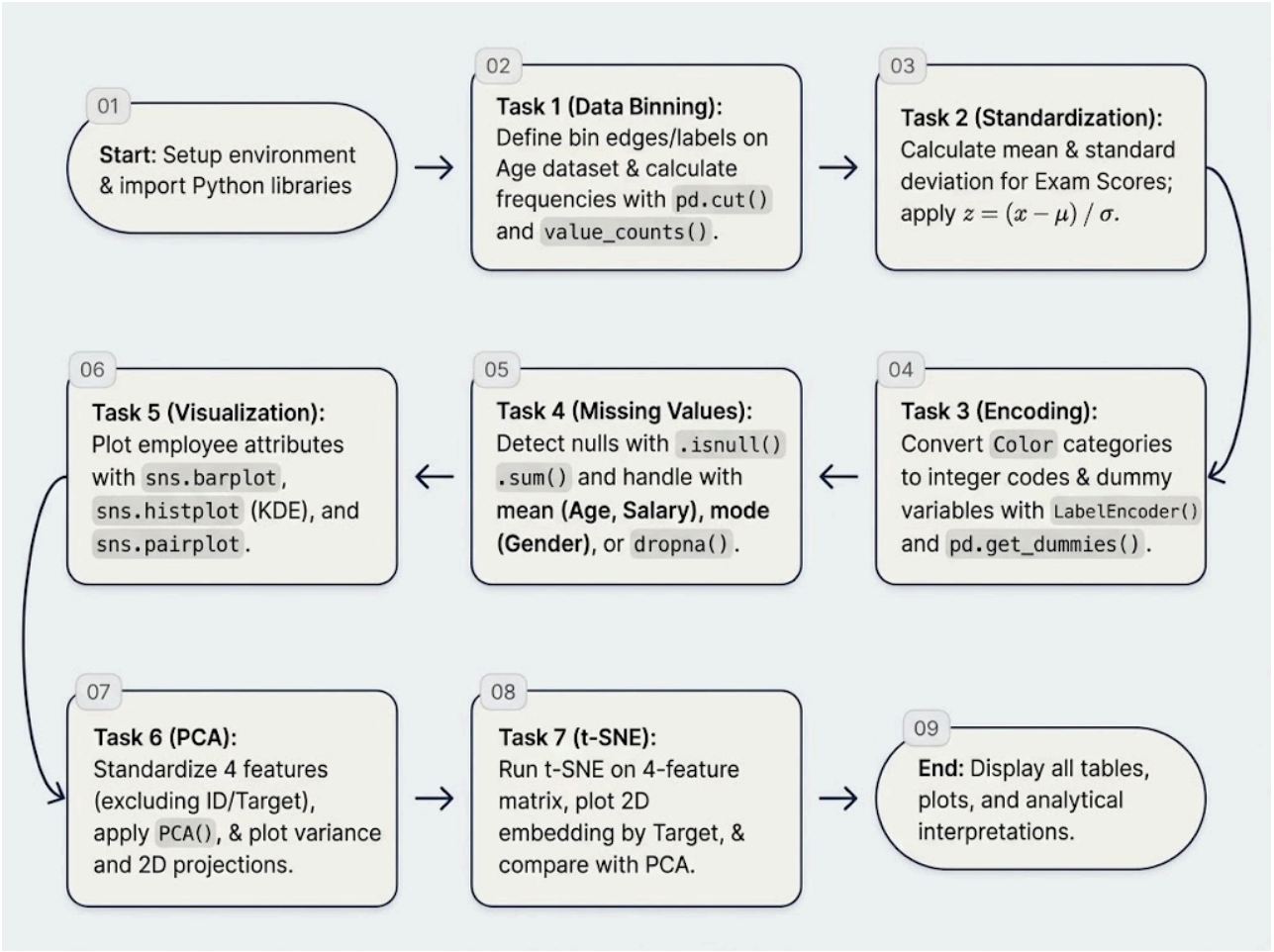


Fig: 1.1: Procedure flow chart.

4. IMPLEMENTATION:

a) Tasks 1 to 3: Binning, Z-Score Standardization, and Encoding:

```
# Task 1: Data Binning
data_task1 = {'ID': [1, 2, 3, 4, 5, 6, 7, 8], 'Age': [12, 25, 47, 33, 56, 80, 21,
39]}
df_task1 = pd.DataFrame(data_task1)
bins = [0, 18, 35, 60, 100]
labels = ['Child/Teen', 'Young Adult', 'Middle-Aged', 'Senior']
df_task1['Age_Group'] = pd.cut(df_task1['Age'], bins=bins, labels=labels)

# Task 2: Standardization Using Z-Score
data_task2 = {'ID': [1, 2, 3, 4, 5, 6], 'Exam_Score': [65, 70, 82, 90, 55, 73]}
df_task2 = pd.DataFrame(data_task2)
mean_score = df_task2['Exam_Score'].mean()
std_score = df_task2['Exam_Score'].std()
df_task2['Z_Score'] = (df_task2['Exam_Score'] - mean_score) / std_score

# Task 3: Categorical Data Encoding
data_task3 = {'ID': [1, 2, 3, 4, 5, 6], 'Color': ['Red', 'Blue', 'Green', 'Red',
'Green', 'Blue']}
df_task3 = pd.DataFrame(data_task3)
df_label_encoded = df_task3.copy()
label_encoder = LabelEncoder()
df_label_encoded['Color_Label'] =
label_encoder.fit_transform(df_label_encoded['Color'])
df_onehot_encoded = pd.get_dummies(df_task3, columns=['Color'], dtype=int)
```

b) Task 4: Handling Missing Values:

```
data_task4 = {'ID': [1, 2, 3, 4, 5], 'Name': ['Alex', 'Ben', 'Chris', 'Dana',
'Eva'],
              'Age': [25, np.nan, 30, 28, 35], 'Gender': ['Male', 'Male', np.nan,
'Female', 'Female'],
              'Salary': [50000, 60000, 70000, np.nan, 80000]}
df_task4 = pd.DataFrame(data_task4)
```

```
df_imputed = df_task4.copy()
df_imputed['Age'] = df_imputed['Age'].fillna(df_imputed['Age'].mean())
df_imputed['Salary'] = df_imputed['Salary'].fillna(df_imputed['Salary'].mean())
df_imputed['Gender'] = df_imputed['Gender'].fillna(df_imputed['Gender'].mode()[0])
```

c) Task 5: Data Visualization Using Seaborn:

```
data_task5 = {'ID': [1, 2, 3, 4, 5, 6, 7, 8],
              'Gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female',
'Male', 'Female'],
              'Age': [25, 30, 22, 35, 40, 28, 32, 45], 'Salary': [50000, 60000,
45000, 70000, 80000, 55000, 62000, 90000],
              'Experience': [2, 5, 1, 8, 10, 3, 6, 12]}
df_task5 = pd.DataFrame(data_task5)
# Plots generated using sns.barplot(), sns.histplot(), and sns.pairplot()
```

d) Tasks 6 & 7: PCA and t-SNE:

```
data_task6_7 = {'ID': range(1, 16), 'Age': [29, 34, 37, 41, 45, 31, 39, 43, 48, 50,
52, 55, 58, 47, 60],
                'RestingBP': [120, 118, 125, 130, 128, 115, 122, 132, 135, 140,
138, 145, 150, 142, 155],
                'Cholesterol': [180, 190, 205, 210, 220, 175, 198, 215, 235, 245,
230, 250, 265, 240, 280],
                'MaxHR': [175, 168, 165, 160, 155, 180, 170, 158, 150, 145, 148,
140, 138, 152, 130],
                'Target': [0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1]}
df_pca = pd.DataFrame(data_task6_7)
feature_cols = ['Age', 'RestingBP', 'Cholesterol', 'MaxHR']
X = df_pca[feature_cols]
y = df_pca['Target']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# PCA
pca_full = PCA()
```

```
X_pca_full = pca_full.fit_transform(X_scaled)

# t-SNE
tsne = TSNE(n_components=2, perplexity=5, random_state=42)
X_tsne = tsne.fit_transform(X_scaled)
```

5. OUTPUT

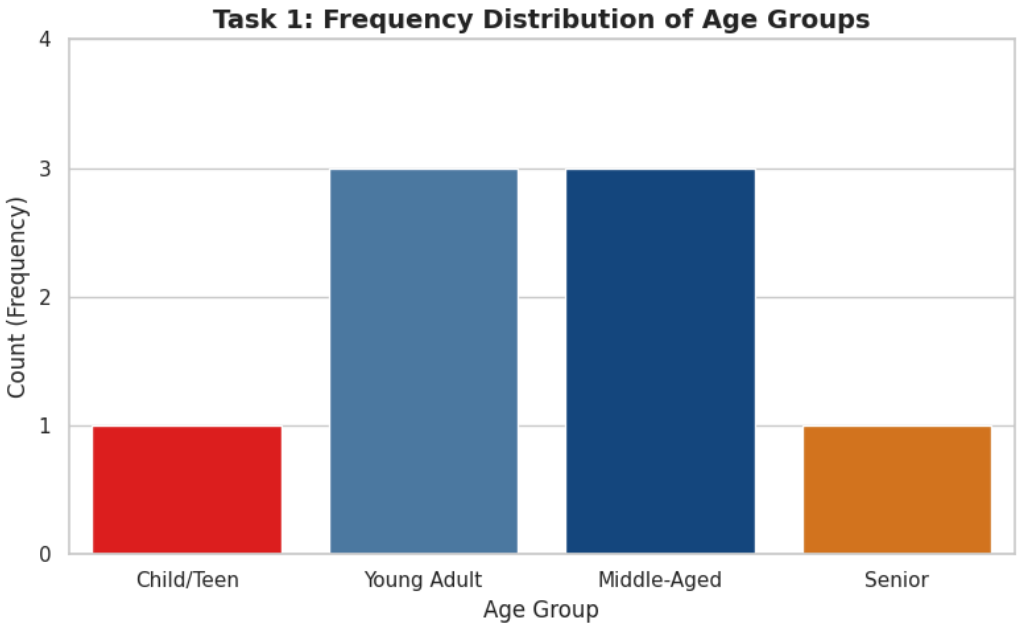


Fig-1.2: Result of (a) task 1.

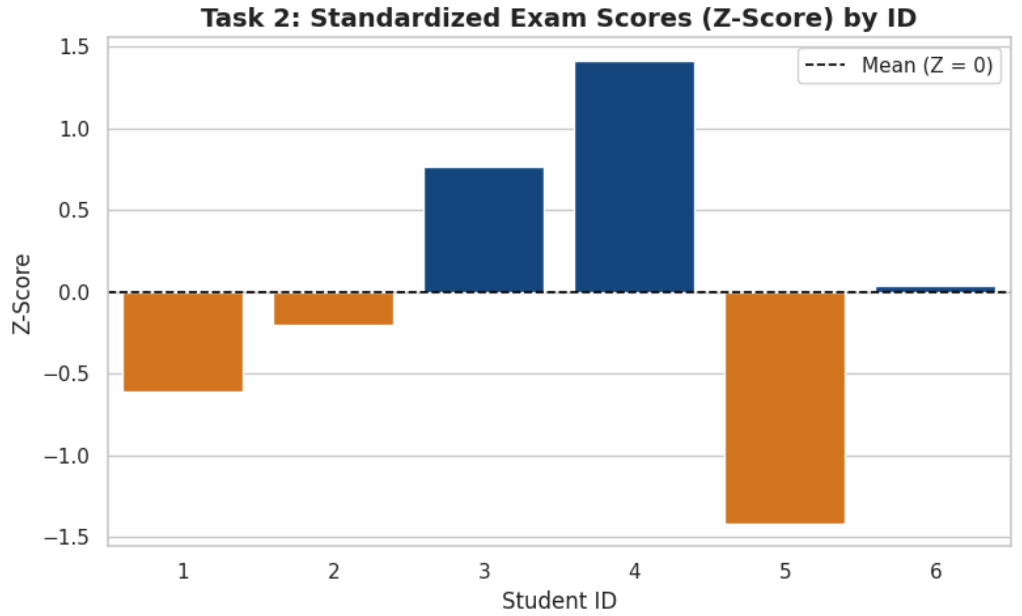


Fig-1.3: Result of (a) task 2.

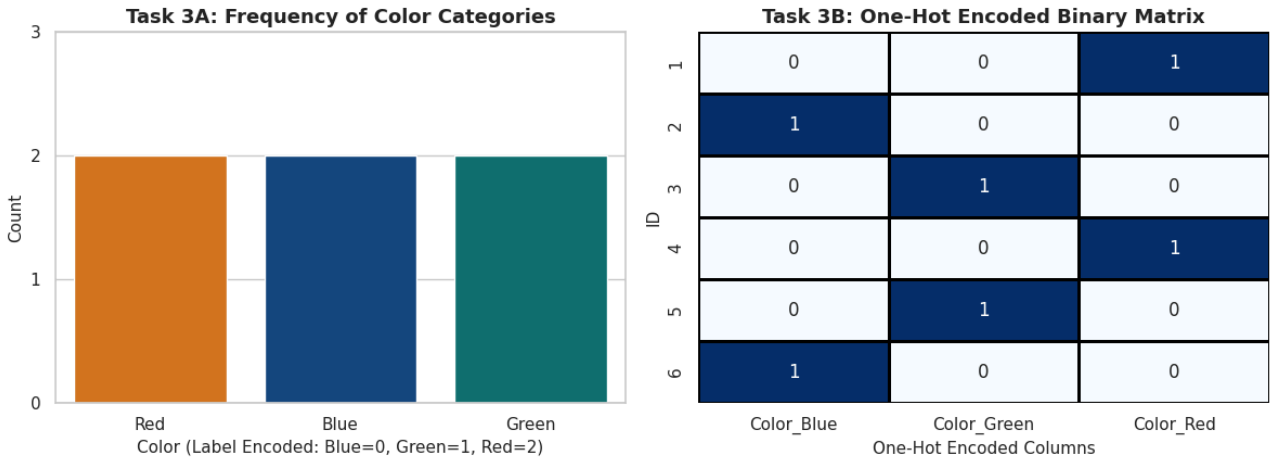


Fig-1.4: Result of (a) task 3.

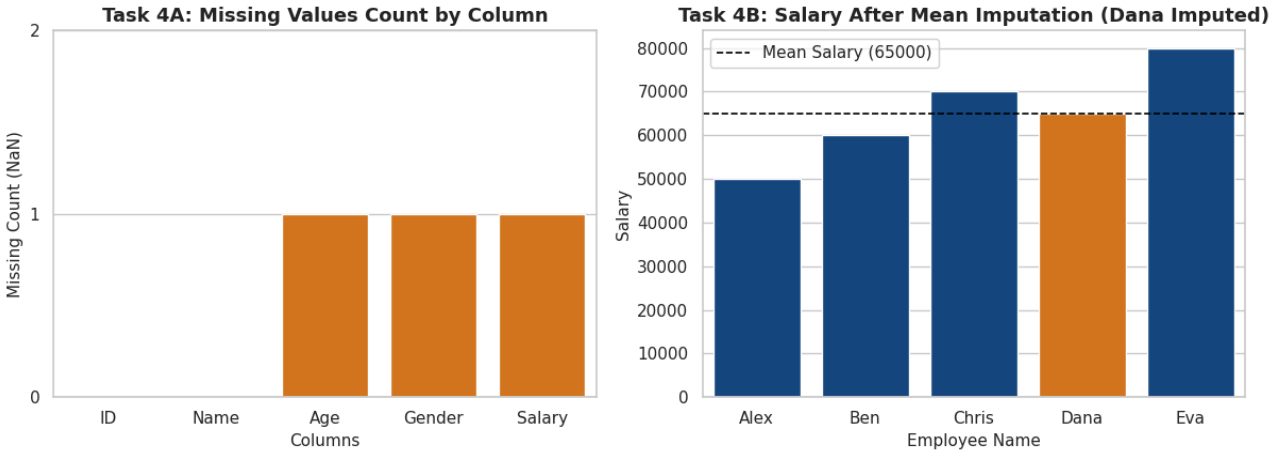


Fig-1.5: Result of (b) task 4.

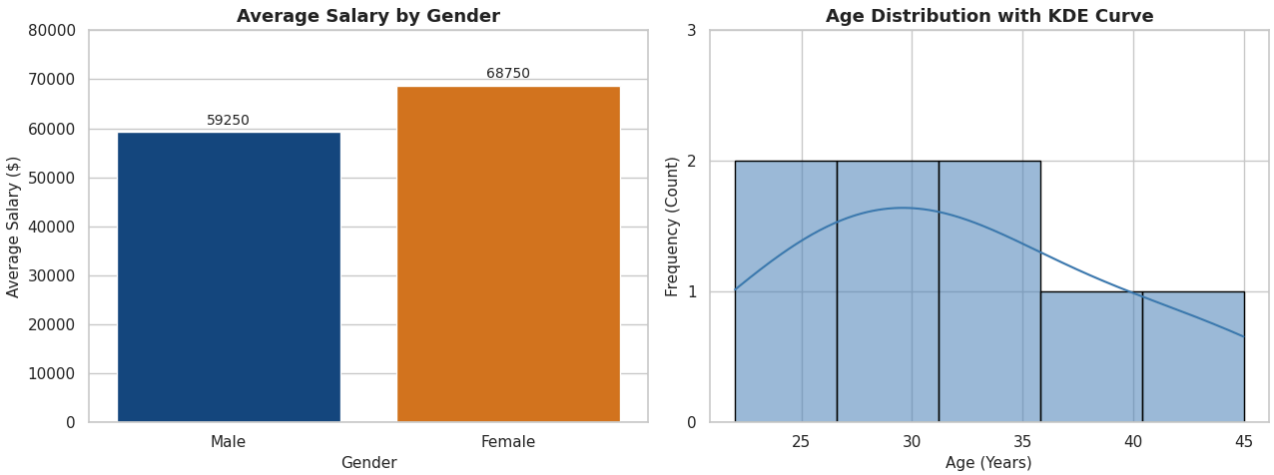


Fig-1.6: Result of (c) task 5.

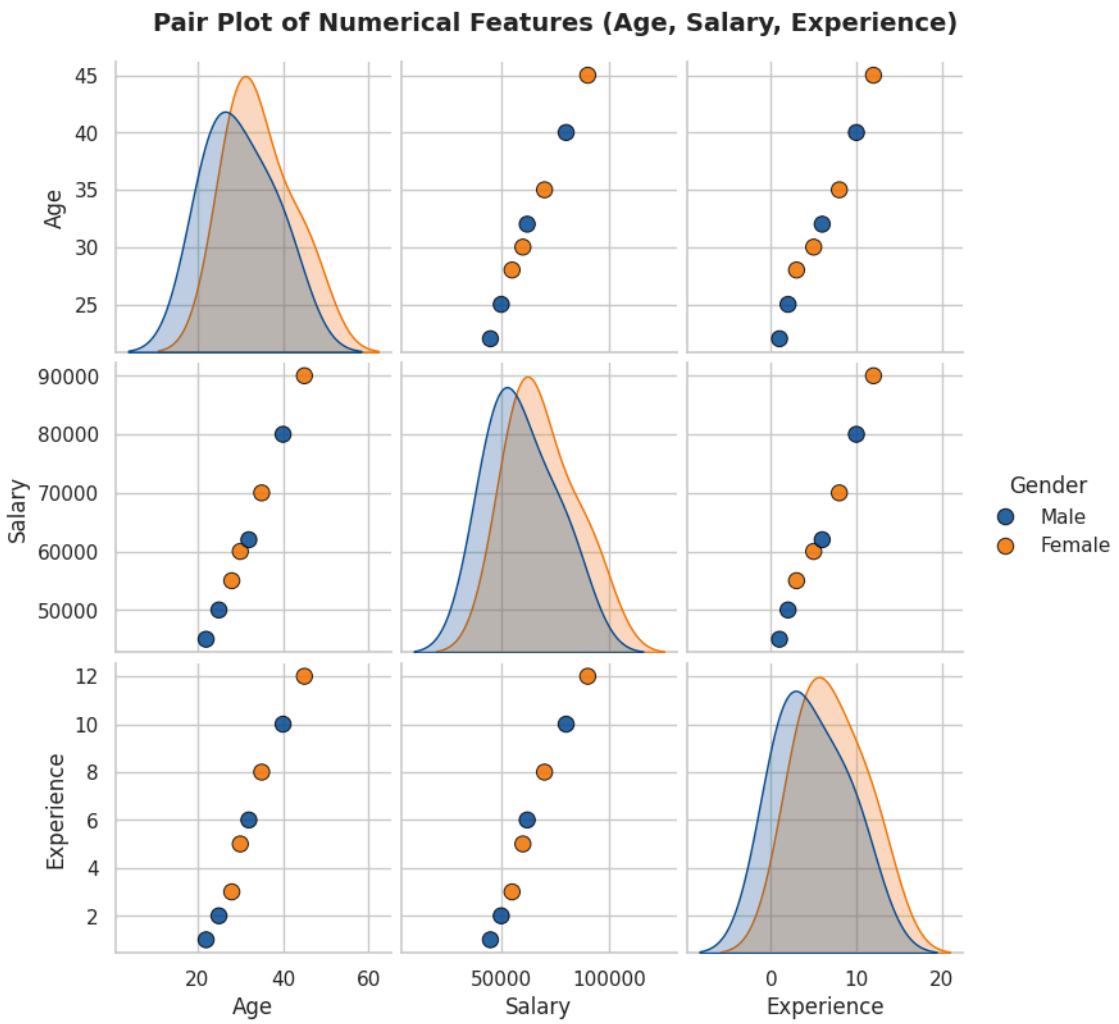


Fig-1.7: Result of (c) task 5.

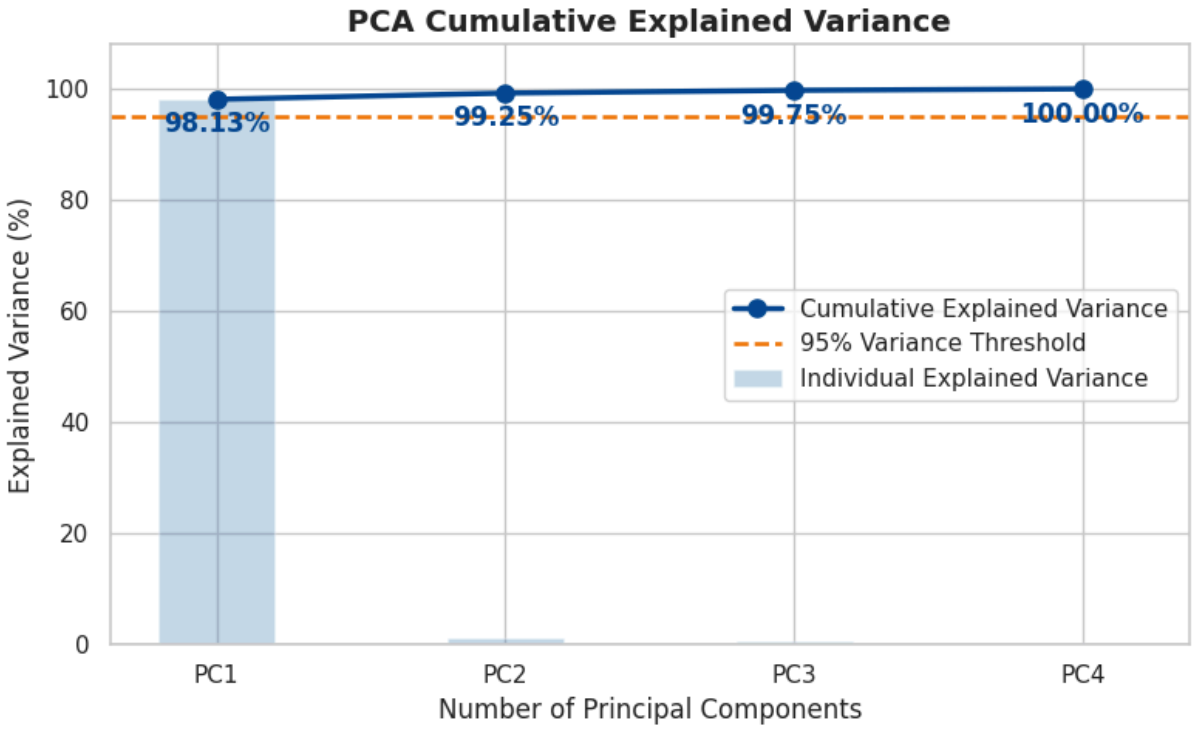


Fig-1.8: Result of (d) task 6.

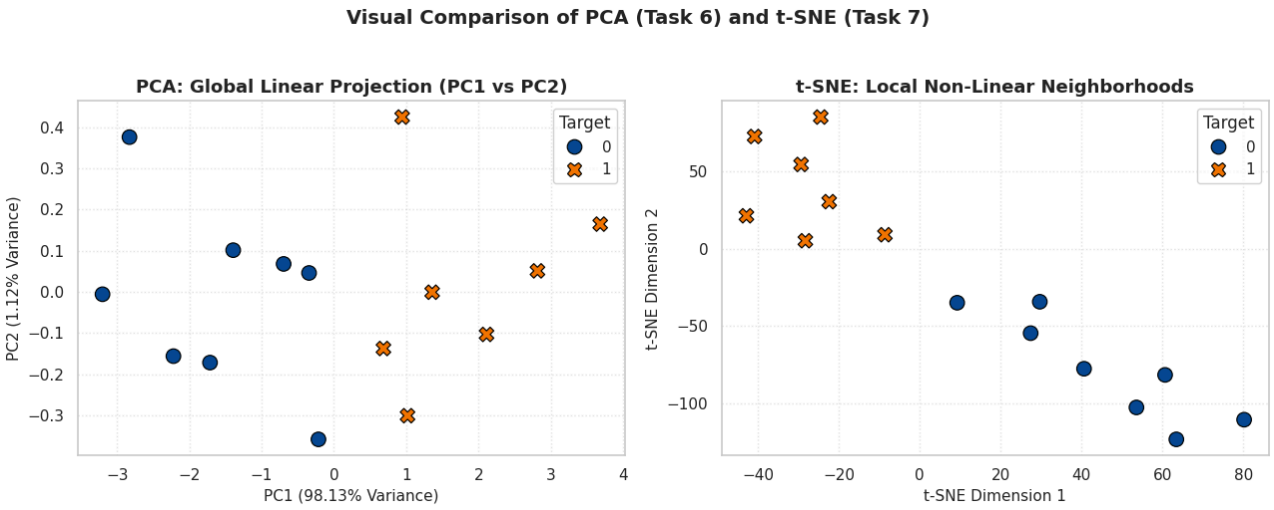


Fig-1.9: Result of (d) task 6 and 7.

6. ANALYSIS AND DISCUSSION

Task 1: Data Binning

The continuous Age values, ranging from 12 to 80, were divided into four life-stage categories using the pd.cut() function: Child/Teen, Young Adult, Middle-Aged, and Senior. The frequency distribution showed that Young Adult and Middle-Aged were the most represented categories, with three individuals in each group, while Child/Teen and Senior had one individual each. This categorization made the age distribution easier to interpret and helped summarize continuous data into meaningful groups.

Task 2: Z-Score Standardization

Z-score standardization was applied to the Exam_Score dataset, which had a mean of 72.50 and a standard deviation of 12.37. The transformation expressed each score in terms of its distance from the mean, resulting in standardized values centered around zero. IDs 3, 4, and 6 had positive Z-scores because their scores were above the mean, whereas IDs 1, 2, and 5 had negative Z-scores. ID 4 recorded the highest positive Z-score (+1.41), while ID 5 had the lowest Z-score (−1.41). This method makes it easier to compare individual scores and understand their relative positions within the dataset.

Task 3: Categorical Data Encoding

The nominal feature Color, containing equal numbers of Red, Blue, and Green observations, was encoded using Label Encoding and One-Hot Encoding. Label Encoding assigned integer values to the categories alphabetically: Blue = 0, Green = 1, and Red = 2. Although this approach is simple and memory-efficient, it may introduce an unintended sense of order among categories that have no natural ranking. In contrast, One-Hot Encoding created three separate binary columns, allowing each color to be represented independently. Therefore, One-Hot Encoding is generally more suitable for nominal features when an artificial numerical relationship between categories should be avoided.

Task 4: Handling Missing Values and the Appropriateness of dropna()

The dataset contained one missing value each in Age, Gender, and Salary. Mean imputation was used to fill the missing numerical values, assigning 29.5 to Ben's Age and 65,000 to Dana's Salary. The missing Gender value for Chris was replaced with the mode, Female. These methods allowed the dataset to remain complete while retaining all five records.

The comparison with dropna() demonstrated the importance of selecting an appropriate missing-value treatment strategy. Removing rows from a small dataset can result in substantial information loss. In this case, applying dropna(axis=0) would remove 60% of the records, leaving only two rows. Imputation was therefore a more appropriate choice for this dataset. Row removal may be reasonable when missing records represent a negligible proportion of a sufficiently large dataset, while column removal may be considered when a feature contains extensive missing data and has limited analytical value. These decisions should ultimately depend on the dataset's size, the missing-data pattern, and the importance of the affected features.

Task 5: Data Visualization Using Seaborn

Seaborn was used to explore employee data through bar plots, histograms, and pair plots. The bar plot showed that female employees had a higher average salary of 68,750 compared with 59,250 for male employees. The female group also had greater average work experience, at 7.0 years compared with 4.75 years. These observations suggest a possible relationship between work experience and salary within this dataset, although further analysis would be required to establish the strength and significance of that relationship.

The histogram illustrated the distribution of employee ages, which ranged from 22 to 45 years, with a noticeable concentration in the late twenties and mid-thirties. The pair plot further revealed positive relationships among Age, Experience, and Salary. Together, these visualizations provided an accessible overview of the dataset and helped identify patterns that could be investigated in greater detail.

Task 6: Dimensionality Reduction Using PCA

Principal Component Analysis (PCA) was applied to four standardized numerical features: Age, RestingBP, Cholesterol, and MaxHR. The results showed that the first principal component (PC1) explained 97.96% of the total variance, while the second principal component (PC2) explained an additional 1.43%. Combined, the first two components accounted for 99.39% of the variance.

Since PC1 alone exceeded the 95% explained-variance threshold, a single principal component was sufficient to retain most of the variation in the transformed dataset. This substantial concentration of variance in PC1 suggests that the standardized features may contain considerable redundancy or share a dominant underlying pattern. The PC1-versus-PC2 scatter plot also showed a clear separation between Target = 0 and Target = 1, with the two classes appearing on opposite sides of the horizontal axis. However, this visual separation should be verified using appropriate classification and validation methods before drawing conclusions about predictive performance.

Task 7: Dimensionality Reduction Using t-SNE and Comparison with PCA

The t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm was used to project the four standardized features into a two-dimensional space with n_components=2, perplexity=5, and random_state=42. The resulting visualization showed two distinct clusters corresponding to Target = 0 (IDs 1–8) and Target = 1 (IDs 9–15). This representation made the local grouping patterns within the dataset easier to observe.

Although both PCA and t-SNE were used to visualize the data in a lower-dimensional space, they serve different purposes. PCA is a linear dimensionality-reduction technique that identifies directions of maximum variance and provides interpretable explained-variance ratios. In this experiment, PC1 captured most of the total variance. In contrast, t-SNE is a nonlinear technique designed primarily to preserve local neighborhood relationships, making it useful for exploring potential clusters in complex datasets. Its axes do not have direct interpretations in terms of explained variance, and the distances between well-separated clusters should not automatically be treated as meaningful measures of global dissimilarity. Therefore, PCA is useful for understanding overall variance and reducing dimensionality, whereas t-SNE is particularly helpful for visualizing local structures and potential clusters.

7. SUMMARY

This laboratory experiment covered seven fundamental data mining tasks using Python, including data preprocessing, encoding, missing-value handling, visualization, and dimensionality reduction. Z-score standardization, Label Encoding, and One-Hot Encoding were applied to prepare data for analysis, while mean and mode imputation were used to handle missing values.

Seaborn visualizations helped identify patterns in employee data. PCA and t-SNE were used to explore multidimensional data, with the first principal component explaining 97.96% of the total variance. Overall, the experiment provided practical knowledge of essential data mining techniques and their importance in effective data analysis and machine learning.

8. GITHUB REPOSITORY:

Link: https://github.com/pro382r/GUB-Academic/tree/main/DML/lr1

9. COLAB FILE:

- Link: DML_223002038_lr1.ipynb